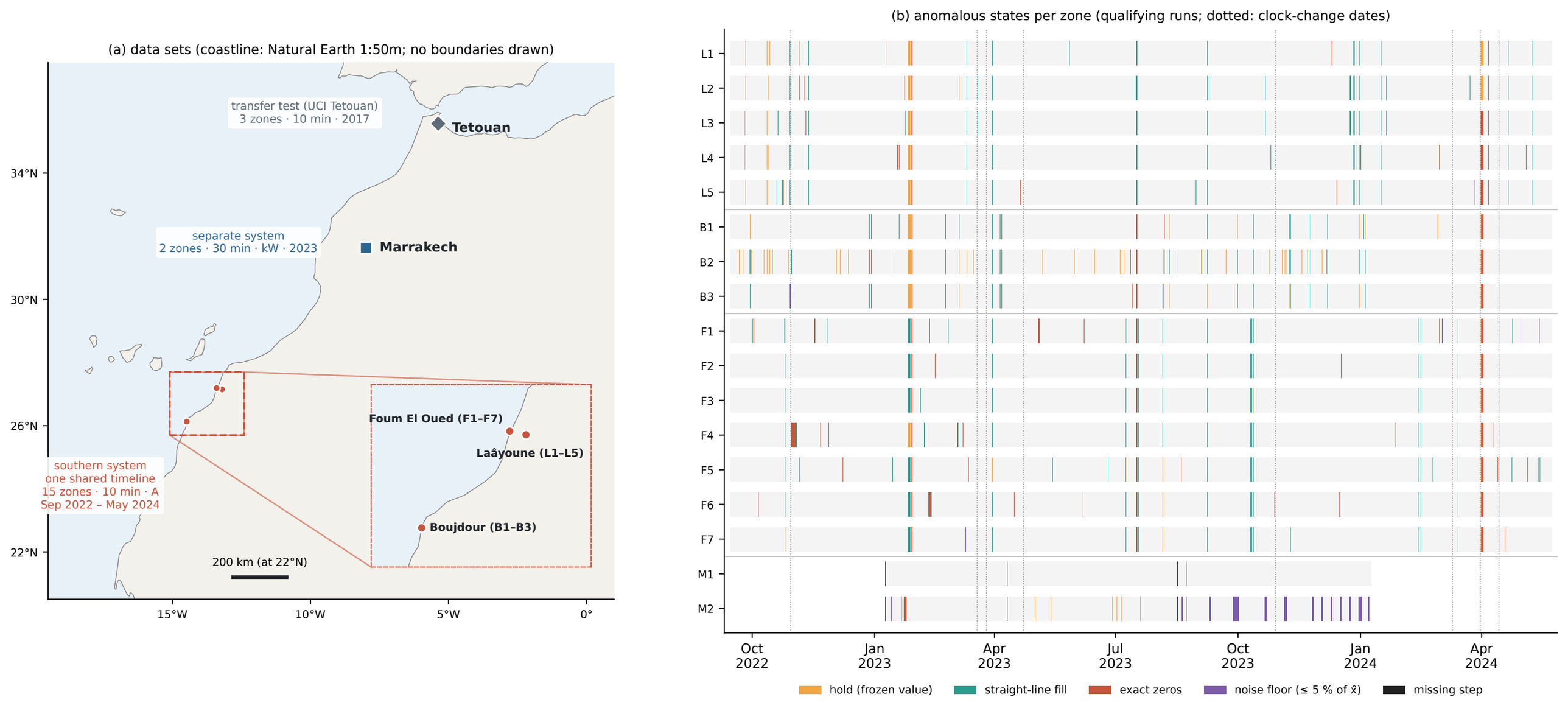

(a) data sets (coastline: Natural Earth 1:50m; no boundaries drawn)
transfer test (UCI Tetouan)
3 zones · 10 min · 2017
Tetouan
separate system
2 zones · 30 min · kW · 2023
Marrakech
southern system
one shared timeline
15 zones · 10 min · A
Sep 2022 – May 2024
Foum El Oued (F1–F7)
Laâyoune (L1–L5)
Boujdour (B1–B3)
200 km (at 22°N)
34°N
30°N
26°N
22°N
15°W
10°W
5°W
0°
(b) anomalous states per zone (qualifying runs; dotted: clock-change dates)
L1
L2
L3
L4
L5
B1
B2
B3
F1
F2
F3
F4
F5
F6
F7
M1
M2
Oct 2022
Jan 2023
Apr 2023
Jul 2023
Oct 2023
Jan 2024
Apr 2024
hold (frozen value)
straight-line fill
exact zeros
noise floor (≤ 5 % of x̂)
missing step